



An efficient nanodiamond-based monolithic foam catalyst prepared by a facile thermal impregnation strategy for direct dehydrogenation of ethylbenzene to styrene

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ABSTRACT

Nanodiamond (ND) has long been recognized as an effective carbocatalyst for synthesizing styrene *via* direct dehydrogenation (DDH). However, the induced drastic pressure drop of its powder form limits its industrial application in heterogeneous catalytic process. In this work, we report a facile hexamethylenetetramine nitrate (HN)-assisted thermal impregnation (HNTI) strategy for fabricating a novel nanodiamond-based monolithic foam (ND/CNT-SiC-ms-HN) catalyst through a two-step approach: One is to soak the carbon nanotube-modified SiC foam (CNT-SiC) with the slurry composed of HN, KCl, LiCl, and dispersed ND, and the other is to heat the slurry-soaked CNT-SiC (ND-HN-KCl-LiCl/CNT-SiC) in N₂ atmosphere at 750 °C. The as-synthesized ND/CNT-SiC-ms-HN monolithic foam features the enriched surface ketonic C=O by promoted ND dispersion and O-doping, abundant structural defects, and improved nucleophilicity by N-doping, originating from the promoted ND dispersion by thermal impregnation (TI) in KCl-LiCl molten salt (MS) and the presence of HN in the annealing process. As a result, the ND/CNT-SiC-ms-HN monolithic foam catalyst by HNTI strategy exhibits 1.5 folds higher steady-state styrene rate (5.49 mmol g⁻¹ h⁻¹) associated with 98.4% of styrene selectivity compared to the ND-based monolithic foam catalyst (ND/CNT-SiC). Moreover, the ND/CNT-SiC-ms-HN monolithic foam shows excellent long-term stability for the direct dehydrogenation of ethylbenzene to styrene. This work also comes up with a novel way of preparing other highly-dispersed nanocarbons-based monolithic foam catalysts with promising catalytic performance for diverse transformations.

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Nanocarbon materials, for example, graphene, carbon dots, fullerene, carbon nanofiber (CNF), nanodiamond (ND), carbon nanotube (CNT), featuring unique structure and chemical properties have attracted extensive attention for catalyzing various important chemical processes in the past decades [1–4]. Among which, the powdered ND first produced in the 1960s in explosions features 5–10 nm diamond nanoparticles. The distinctive sp³ inner core and sp² shell hybridized nanostructure coupled with large surface area, high thermal and mechanical stability and easily prepared character make it an ideal catalyst in diverse industrial application [5–7]. ND has shown great potential in alternating the traditional Fe-based catalysts for styrene production for the sake of clean, safe and eco-friendly requirement over the direct dehydrogenation (DDH) of ethylbenzene process, since the industrial K-Fe catalyst suffer from quick deactivation because of carbon deposition and

potassium loss [8,9]. Moreover, the co-feed superheated steam for the purpose of shifting the chemical equilibrium styrene production and eliminating coke deposition results in high energy consumption [10].

Although ND is beyond the shine as a carbon catalyst in the DDH of ethylbenzene reaction, the commercial ND powder produced from the explosion method tend to block the fixed-bed reactors in industrial process, leading to dramatic pressure drop [11]. Meanwhile, because of the existence of surface force and chemical bonding force, individual ND nanoparticles are easy to form an aggregation [12–14]. According to previous reports [15–19], the exposed surface ketonic C=O groups on ND nanosphere are active sites in ethylbenzene dehydrogenation reaction. During this reaction, C=O is converted to C–O–H, and ethylbenzene is activated to styrene. What's more, the aggregation negatively blocks the effective surface sites and significantly depresses the performance in catalytic process. On the other hand, it is still a great challenge in controlling the supra structural properties of ND in the drastic

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